

API Documentation - Student Performance Analytics Portal

This document outlines the REST API routes exposed by the backend Express server, complete with authorization details, input payload specifications, and response status codes.

Base URL

All routes are prefixed by: `/api`

1. Authentication routes (`/api/auth`)

Login User

- **Method & Path:** `POST /api/auth/login`
- **Access:** Public
- **Request Body:**

```
{
  "email": "teacher@school.edu",
  "password": "securepassword123"
}
```

- **Responses:**
 - **200 OK:**

```
{
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLTJ5Ij..."
```

```
"user": {
  "id": "603d2e5b8d23471df4bcde21",
  "name": "Prof. Smith",
  "email": "teacher@school.edu",
  "role": "teacher"
}
```

- **401 Unauthorized:** Invalid credentials.

Register User

- **Method & Path:** POST /api/auth/register
- **Access:** Private (Admin Only)
- **Request Body:**

```
{
  "name": "Jane Doe",
  "email": "student1@school.edu",
  "password": "changeme999",
  "role": "student"
}
```

- **Responses:**
 - **201 Created:**

```
{
  "success": true,
  "message": "User registered successfully",
  "user": {
    "id": "603d2e8b8d23471df4bcde25",
    "name": "Jane Doe",
    "email": "student1@school.edu",
    "role": "student"
  }
}
```

- **400 Bad Request:** Missing/invalid fields or email already exists.
-

2. Student Profile routes (`/api/students`)

Get All Students

- **Method & Path:** `GET /api/students`
- **Access:** Private (Admin, Teacher Only)
- **Response:**
 - **200 OK:**

```
{
  "success": true,
  "students": [
    {
      "_id": "603d2ebb8d23471df4bcde30",
      "studentId": "STU2026001",
      "class": "10-A",
      "userId": {
        "name": "Alex Johnson",
        "email": "alex@school.edu"
      }
    }
  ]
}
```

Create Student Profile

- **Method & Path:** `POST /api/students`
- **Access:** Private (Admin Only)
- **Request Body:**

```
{
  "userId": "603d2e8b8d23471df4bcde25",
```

```
{
  "studentId": "STU2026001",
  "class": "10-A",
  "guardianName": "Robert Johnson",
  "guardianPhone": "+1234567890"
}
```

- **Response:**
 - **201 Created:** Student record initialized.
-

3. Academic Grade routes (/api/grades)

Record Grade

- **Method & Path:** POST /api/grades
- **Access:** Private (Teacher Only)
- **Request Body:**

```
{
  "studentId": "603d2ebb8d23471df4bcde30",
  "subjectId": "603d2efb8d23471df4bcde45",
  "marksObtained": 85,
  "maxMarks": 100,
  "examType": "Midterm",
  "semester": "Fall 2026"
}
```

- **Response:**
 - **201 Created:** Grade logged.
-

Get Grades for Student

- **Method & Path:** GET /api/grades/student/:studentId
- **Access:** Private (Admin, Teacher, and Owner Student)
- **Response:**
 - **200 OK:**

```
{
  "success": true,
  "grades": [
    {
      "_id": "603d2f2b8d23471df4bcde50",
      "marksObtained": 85,
      "maxMarks": 100,
      "examType": "Midterm",
      "semester": "Fall 2026",
      "subjectId": {
        "name": "Mathematics",
        "code": "MATH101"
      }
    }
  ]
}
```

4. Attendance routes (`/api/attendance`)

Log Attendance

- **Method & Path:** `POST /api/attendance`
- **Access:** Private (Teacher Only)
- **Request Body:**

```
{
  "studentId": "603d2ebb8d23471df4bcde30",
  "date": "2026-06-24",
  "status": "Present",
  "remarks": "On time"
}
```

- **Response:**
 - **201 Created:** Attendance record logged.

Get Attendance History

- **Method & Path:** GET /api/attendance/student/:studentId
- **Access:** Private (Admin, Teacher, and Owner Student)
- **Response:**
 - **200 OK:**

```
{
  "success": true,
  "attendance": [
    {
      "date": "2026-06-24T00:00:00.000Z",
      "status": "Present",
      "remarks": "On time"
    }
  ]
}
```